

08/09

6th topic

Conditional Statement &

→ Conditional Statement & allow us to make decision in code. They check condition (expression that results in true or false) and executing different blocks of code accordingly.

Types of Conditional Statements in Python

1. if Statement - execute the block of code only if the condition is true.
2. if-else Statement - Run one block if the condition is true, otherwise run another block.
- 3) if... elif... else Statement (or) ladder - Multiple conditions checked one by one.
- 4) Nested if → An if statement inside another if statement.

1) if Statement -

Syntax:

if (Condition) :
Statement

Example:-

01)

age = 20

if age >= 18:

Print ("you are eligible to vote")

Here, the condition age >= 18 is checked

If it is True, the message will be printed.

If it is False, nothing happens (since there is no else)

if u miss it won't print that line
& just go to next cell

02)

age = int (input ("enter your age:"))

Country = input ("enter your country:")

if (age >= 18 and country == "India"):

Print ("eligible for voting")

o/p: Enter your age: 18

Enter your country: India

eligible for voting

2) if... else Statement:-

Syntax:- if (Condition) :

Statements # block of code if condition is True

else:

Statements # block of code if condition is False.


```
Ex:- age = int(input("enter your age :"))  
Country = input("enter your country :")  
if (age >= 18 and country == "India"):
```

```
    print("eligible for voting")  
else:
```

```
    print("not eligible for voting")
```

Note:- Python is also Case-Sensitive
if you type ~~Ind~~ india instead of India u
will get not eligible.

```
age = int(input("enter your age :"))  
Country = input("enter your country :")
```

```
Country = Country.lower()
```

```
print(Country)
```

```
if (age >= 18 and country == "india"):
```

```
    print("eligible for voting")
```

```
else:  
    print("not eligible for voting")
```

```
country = country.lower()
```

→ Converts whatever the User typed into lowercase
→ this is done so that India, INDIA, or india all
become india.

→ Makes checking easier

```
Ex:- "India".lower() → "india"
```

Note:- In if condition if you type country india
in lowercase then when u updating the variable
u should write country.lower() then ~~to~~
whatever User type means uppercase or lowercase if
convert to lowercase

country = input ("enter your country :")
country = country.lower()

In Python, you are not declaring a variable twice :
you are just reassigning (updating) its value.

First line

country = input ("enter your country :")

- Takes input from the user
- Suppose you type : India
- Now country = "India"

Second line

country = country.lower()

- Here you are saying

Take the current value of country → make it lowercase →
Store it back into the same variable country.

- "India".lower() → "india"
- Now, country = "india"

- First time → variable is created and stores the user input.
- Second time → Variable is updated with the lowercase version of that input.

This is called variable reassignment.

(You are not declaring twice, you are just replacing the old value with new one.)

8) if... elif... else

Syntax:-

if (condition):

Statements of condition 1

else:

if (condition 2):

Statements of condition 2

else:

if (condition 3):

Statements of condition 3

else:

else block statements

Note:- One if condition must & should have else condition

instead.

if (condition):

Statements of condition 1

elif (condition 2):

Statements of condition 2

elif (condition 3):

Statements of condition 3

else:

else block statements

elif

else & if both
combined

else statement is
replaced by if

if condition 1:

block of code if condition 1 is True

elif condition 2:

block of code if condition 2 is True

elif condition 3:

block of code if condition 3 is True

else:

block of code if none of the above conditions are
True

Note:- Only one block will execute

Python checks conditions from top to bottom

As soon as one condition is True, it runs that
block & skips the rest

```
marks = int(input("Enter your marks: "))
```

```
if marks >= 90:
```

```
    print("Grade: A")
```

```
elif marks >= 75:
```

```
    print("Grade: B")
```

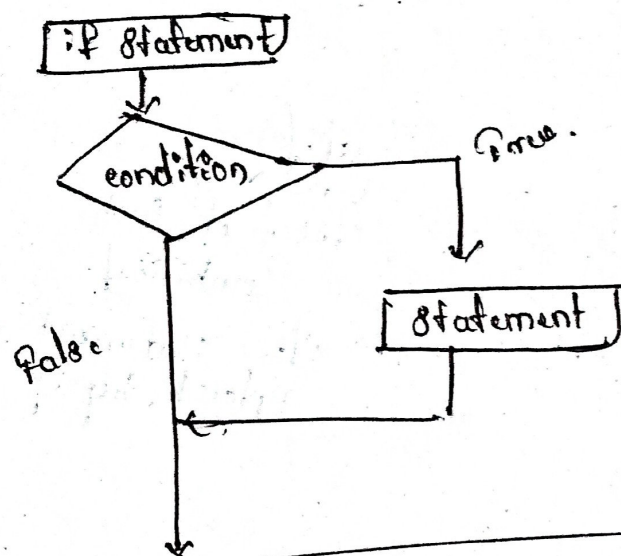
```
elif marks >= 50:
```

```
    print("Grade: C")
```

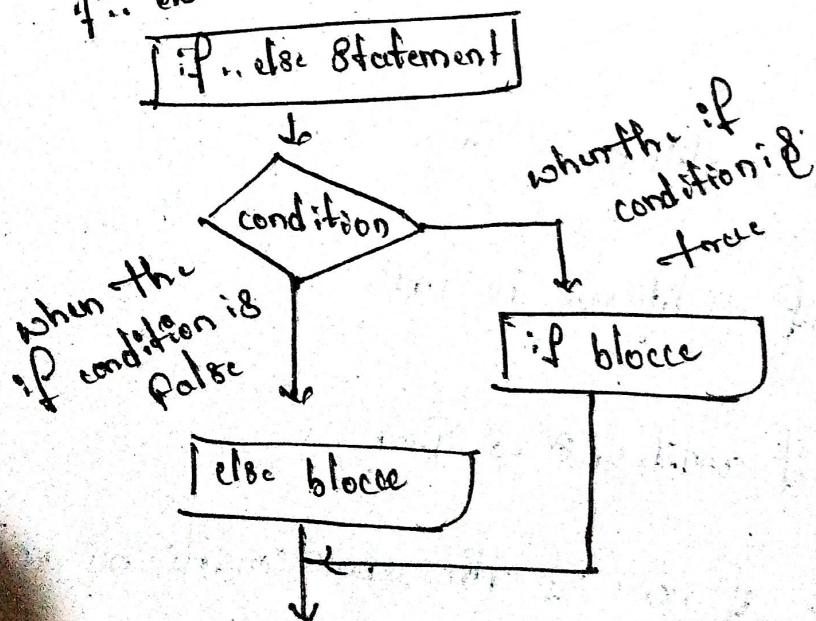
```
else:
```

```
    print("Grade: Fail")
```

Block diagram for if statement



if... else statement



if.. elif.. else statement

